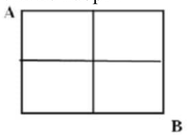
45. Study the following Figure

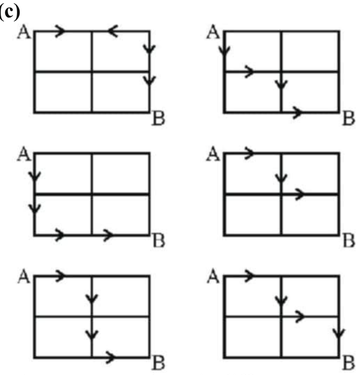
A person goes from A to B always moving to the right or downwards along the lines. How many different routes can he adopt ?



Select the correct answer from the codes given below :

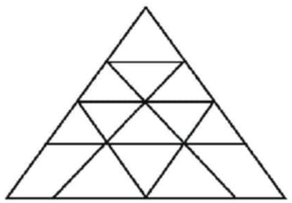
(a) 4 (b) 5
(c) 6 (d) 7

Ans. (c)



Hence, A person can adopt six different routes from A to B.

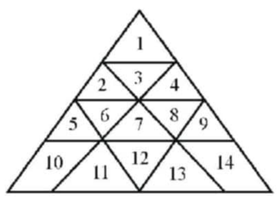
46. Consider the following figure and answer the item that follows :



What is the total number of triangles in the above grid ?

(a) 27 (b) 26
(c) 23 (d) 22

Ans. (c)



∵ Triangle formed by one-digit = 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 13

∴ No. of triangle formed by one-digit = 12

∵ Triangle formed by three-digit = (5, 10, 11) (9, 13, 140

∴ No. of triangle form by three digit = 2

∵ Triangle formed by four digit = (6, 7, 8, 12) (7, 11, 12, 13) (1, 2, 3, 4) (2, 5, 6, 7) (4, 7, 8, 9)

∴ No. of triangle by four digit = 5

triangle formed by others digit ⇒

(1, 2, 3, 4, 5, 6, 7, 8, 9)
(2, 5, 6, 7, 10, 11, 12, 13)
(4, 7, 8, 9, 11, 12, 13, 14)
(1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14)

∴ No. of triangular formed other digit = 4

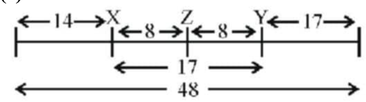
Hence,

Total no. of triangle in grid
= 12 + 2 + 5 + 4
= 23

47. In a queue, Mr. X is fourteenth from the front and Mr. Y is seventeenth from the end, while Mr. Z is exactly in between Mr. X and Mr. Y . If Mr. X is ahead of Mr. Y and there are 48 persons in the queue, how many persons are there between Mr. X and Mr. Z ?

(a) 6 (b) 7
(c) 8 (d) 9

Ans. (c)



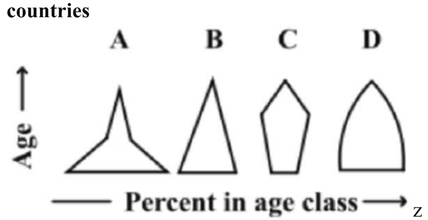
The no. of persons between Mr. X and Mr. Y

⇒ 48 – (14 + 17)
= 48 – 31
= 17

∴ There are 8 persons between Mr. X and Mr. Z.

BASIC NUMERICAL ABILITY

48. Consider the four age pyramids given below namely A, B, C and D representing four different countries

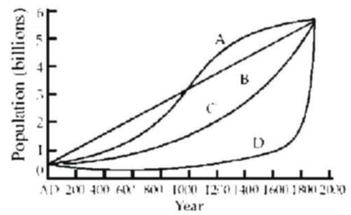


Which one of them indicates the declining population?

(a) A (b) B
(c) C (d) D

Ans. (c) Option 'C' indicates the declining population

49. The following figure has four curves namely A, B,C and D. Study the figure and answer the item that follows.



Which curve indicates the exponential growth?

(a) A (b) B
(c) C (d) D

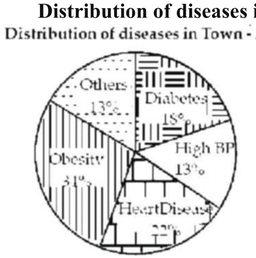
Ans. (c) Curve C indicates the exponential growth.

Direction for the following 2 (two) items :

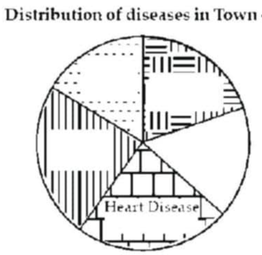
The following pie charts show the break-up of disease categories recorded in the patients from two towns. Town A and Town B. Pie charts plot the disease categories as percentage of the total number of patients.

Based on these, answer the two items that follow the charts.

Distribution of diseases in Town-A



Distribution of diseases in Town-B



50. Which of the two town has a higher number of persons with Diabetes ?

(a) Town /शहर A
(b) Town /शहर B
(c) Same in Town A and Town B
(d) No inference can be drawn

Ans. (d) No inference can be drawn.

∵ Given the disease categories as percentage of the total number of patient but not given that number of patients.

51. What can we say about persons with more than one disease from these graphs ?

(a) There are likely to be persons with more than one disease in Town A
(b) There are likely to be persons with more than one disease in Town B
(c) There are likely to be persons with more than one disease in both Towns A and B
(d) No Inference can be drawn.

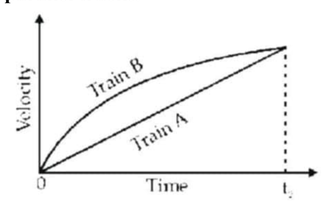
Ans. (b) ∵ Town A = (18 + 13 + 22 +34 + 13)% ⇒ 100%

and town B = (24 + 19 + 30 + 28 + 20)% = 121%

Hence,

There are likely to be person with more than one disease in town B.

52. Consider the following Velocity-Time Graph. It shows two trains starting simultaneously on parallel tracks.



With reference to the above graph, which one of the following statements is not correct?

(a) Train B has an initial acceleration greater than that of Train A.
(b) Train B is faster than Train A at all times
(c) Both trains have the same velocity at time T_0
(d) Both trains travel the same distance in time t_0 units

Ans. (d) The velocity of an object is the rate of charge of its position with respect to a frame of reference and is a function of time.

$$\bar{v} = \frac{\Delta d}{\Delta t}$$

Acceleration is the rate of velocity.

$$\bar{a} = \frac{v - v_0}{t_1 - t_2}$$

$$\bar{a} = \frac{\Delta v}{\Delta t}$$